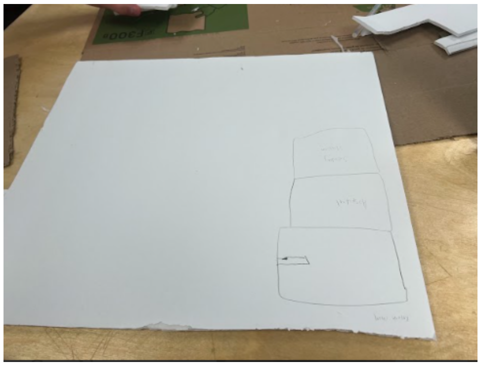
Name: I Forgot Again Team **Date:** 4/15/24 **Class:** Engineering

MyDesign Portfolio

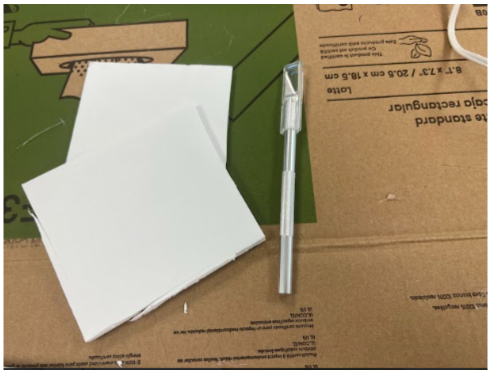
Element G: Construction of a Testable Prototype

Date: 4/15/24

PROTOTYPE:



1. Prepare foam for cutting and mark where to cut.



2. Cut the foam to the appropriate size.

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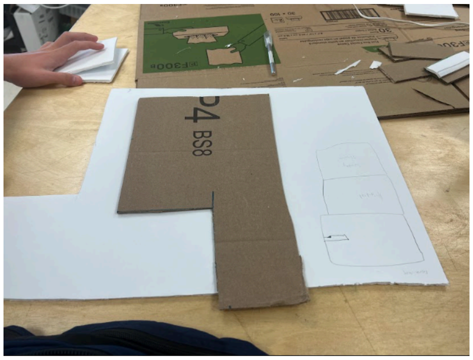
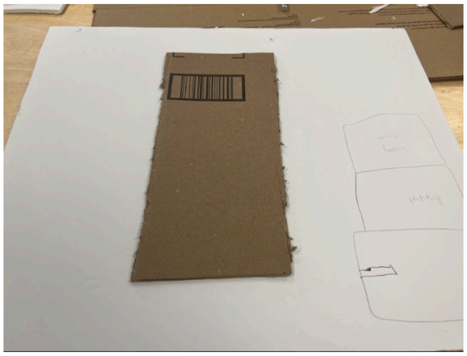
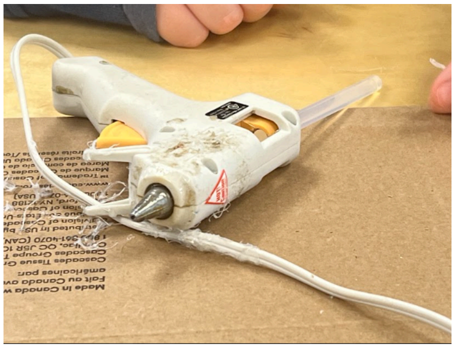
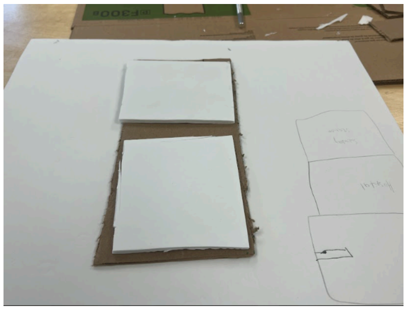


3. Prepare cardboard for cutting and mark where to cut.



4. Cut the cardboard.

5. Prepare the hot glue gun and use hot glue to attach the foam to the cardboard.

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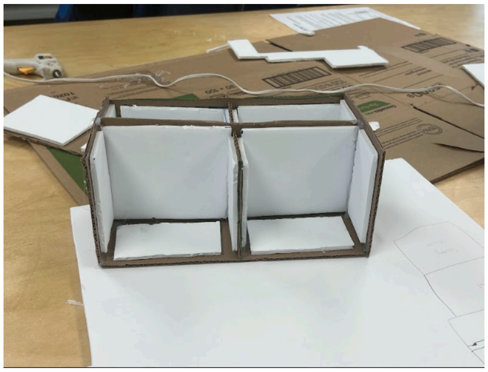
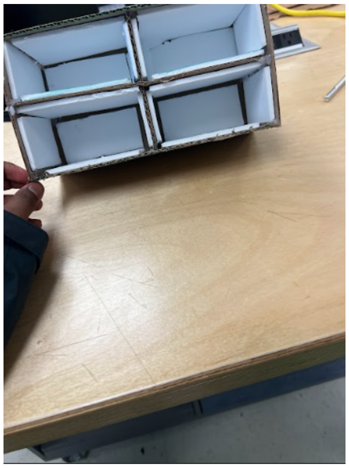


6. Hot glue cardboard together in between the foam.



7. END

	Details	Tools and Manufacturing Methods Used
Step 1	Find and measure out pieces of foam for our design. Some are squares, some are rectangles. All must be the right size.	We used a ruler, pencil, and foam to measure our design properly.

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Step 2	Cut out the foam squares for the prototype. There are 12 smaller rectangle ones and 8 larger square ones for a total of 20 foam supports.	We used a cutting knife and the foam to cut our foam to fit.
Step 3	Find and measure out cardboard pieces for the prototype. The design is a rectangular prism with no lid and a 4-way divider in the middle.	We used a ruler, pencil, and cardboard to measure our prototype. To create the 4-way divider, you can make 2 separate peaces that interlock with each other. One piece should be the same length as the prism and one should be the same width. Then, make a cut in both where they can interlock.
Step 4	Cut the cardboard using the knife to create 7 different pieces, 5 apart from the prism and 2 to create the divider.	We used a cutting knife and our cardboard.
Step 5	Properly use the hot glue gun to glue the foam boards onto the cardboard.	We used a glue gun, our foam, and our cardboard.
Step 6	After step 5, use the hot glue gun to glue the new boards together to form the prism. Note that the middle divider should be placed	We used a glue gun and our new boards.

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	first against the floor in order for it to center the rest of the walls. Glue every edge that each board connects too for best results.	
Step 7	END	END

Our design has had multiple changes from the blueprints. We had to change our measurements and scale down our design because the original size wouldnt fit in the car console. Next, we went from 6 compartments to 4 compartments because with the new size 6 compartments wouldn't fit.

FINAL ITERATION:

Date: 4/19/24

We added a lid with latches to the top.

	Details	Tools and Manufacturing Methods Used

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Step 1	Follow all steps from the original prototype.	Follow all steps from the original prototype.
Step 2	Measure out a base of cardboard that is the same dimensions as the base of the prototype, 2 latches that are .5 inches by 2 inches, and 4 sides that are 2 inches tall that otherwise match the dimensions of the prototype.	We used a ruler, cutting knife, hot glue gun, and cardboard. Use the same construction methods from the prototype.
Step 3	Glue the rectangular peaces to create an open rectangular prism.	We used a hot glue gun and cardboard.
Step 4	Glue the latches onto the new open rectangular prism.	We used a hot glue gun and cardboard.
Step 5	Glue the latches to the long side of the prototype (Make sure the lid can open and close properly).	We used a hot glue gun and cardboard.
Step 6	END	END

We made this iteration because the new lid will keep our items safe and will be more convenient for the driver.

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Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item
cardboard	The remains of a sheet of cardboard already cut in a previous project	Supplied by producer	N/A	N/A	N/A
Hot glue stick	A 2-inch long cylinder of solid glue	Supplied by producer	N/A	N/A	N/A
Sheet of foam	A thin sheet of foam in a similar condition to the cardboard	Supplied by producer	N/A	N/A	N/A

Detailed sketches, which might include CAD if you have learned it in your class:

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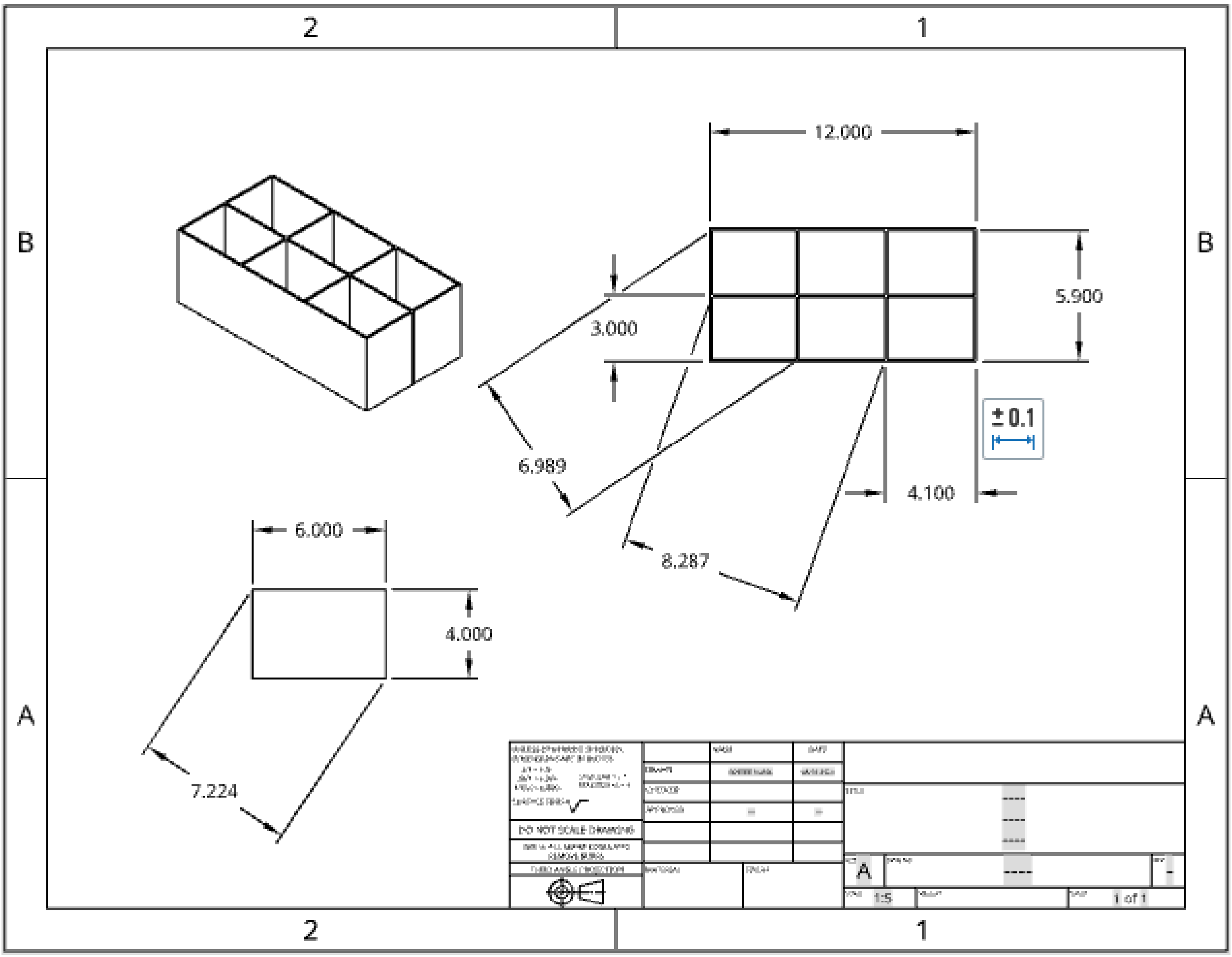

*CAD design is not the final version of our prototype.

Construction of the Final Prototype *(G2)*

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Proper Fit	We will use measurements from a ruler for both the car console and the final prototype to see if our dimensions fit properly.

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Durability	We will use a gravitational equation in physics to determine how high an item has been dropped if their is damage too it from the prototype.
Saftey	We will use the Driver Safety Measurement System (DSMS) Methodology to determine if our driver was distracted while driving. We can also use feedback from the driver to figure out data.

Testing or Modeling of Attributes *(G3, G4)* and Addressing Non-Testable Attributes *(G4)*

Attribute	**Description**	**Describe your prototype's functionality to test this attribute**
Proper Fit	The design should accommodate various car models and measurement configurations to ensure the safety of people and objects.	Our design must fit into the car console for it to work; if it dosen't fit, the prototype dosen't work and loses its functionality.
Durability	The items placed in the prototype must be fully intact during, and after the vehicle is in motion.	Our prototype must keep the items inside intact while in the design. If our items are damaged in any way, then the prototype fails.
Saftey	The driver must be focused on driving and not the location of their objects. In other words, their items	Our designs main goal is to have the driver be more focused on the road. If the design works, the driver can put their items into the

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	must be secure so that the driver can focus.	device without having to worry or be distracted while driving.

Addressing Non-Testable Attributes *(G4)*

Attribute	**Justification for why this attribute cannot be modeled or tested with your prototype**
N/A	N/A
N/A	N/A

Element G: Construction of a Testable Prototype						
	5	4	3	2	1	0
1. Final Prototype Iteration Explanation. The Final Prototype Iteration is ______________.	clearly and *fully* explained	*clearly and adequately explained*	*adequately explained*	*somewhat explained but with gaps*	*only minimally* explained	is *unclear* or is *missing* altogether
2. Construction of the Final Prototype. The Final Prototype Iteration is ______________.	constructed with enough detail to ensure that objective data on *all* or nearly all design requirements could be determined	constructed with enough detail to ensure that objective data on *many or nearly all* design requirements could be determined	constructed with enough detail to ensure that objective data on *some* design requirements could be determined	constructed with enough detail to ensure that objective data on at least a *few* design requirements could be determined	*not* constructed with enough detail to ensure that objective data on at least *one* design requirement could be determined	*not applicable* to any design requirements and/or functional claims

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3. Testing or Modeling of Attributes. ______________ attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are addressed.	*All*	*Most*	*Some*	*A Few*	*No more than one*	*None of the*
4. Addressing non-Testable Attributes. ______________ justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.	A *well* supported	A *generally* supported	An *adequately* supported	*Developing*	*Insufficient*	*No*